



Electronic Data Usage for Equipment Management on Construction Projects

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The introduction of electronic data reporting and tracking on mobile construction equipment has allowed fleet managers to monitor their fleets in 'real time' with an abundance of data flowing in on a daily basis. Unfortunately all of this data can overwhelm a fleet manager and sometimes cause them to micro-manage rather than looking at their overall fleet status. For many fleets, focusing on the basic fleet data can still provide the biggest return in time savings and consistent, correct data coming from their machines.

2.0 Best Practice Description

This Best Practice outlines how focusing on managing a few very basic pieces of electronic data can translate into time efficiencies for and cost savings for an entire equipment department on a construction project where the work and environment changes on a daily basis.

3.0 Implementation Steps

The largest obstacle to implementing electronic data into the fleet management process is to have participation and buy-in from the entire equipment management staff. Everyone needs to be on the same page and understand the benefit to both their job roles as well as the overall fleet benefits. There also needs to be an understanding from the top management level that there is some implementation costs, however those costs will be easily returned in the manpower and equipment cost savings.

4.0 Benefits

This project site is utilizing pieces of Technology, bundled together, each with their own value added traits:

Equipment Location and Runtime

- o **Caterpillar Product Link** is being utilized to aid in managing runtime and utilization reporting.
 - Equip location reported hourly with PL522
 - Equipment runtime and utilization reporting
 - Fuel utilization reporting (burned idle vs. production)
 - Fault code reporting and maintenance plan tracking
 - Geo-fencing parameters and hourly location logger
 - Idling tracking (Idle based on engine RPMs)

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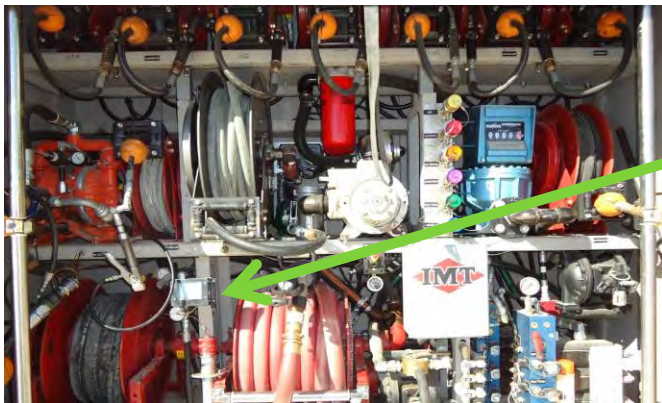
Equipment Fuel Management

o Smart” Lube Trucks

- The smart lube truck is a modern lube truck outfitted with a computer, wireless capabilities, and electronic flow-meters installed on each reel.
- The rear screen displays the list of equipment within a 200 ft radius and the current hour meter reading. This system also records the amount of fluids (fuel/oil) added to the machine and time stamps it with the hour meter.
- There is a tablet in use in the cab of the lube truck, linked to the OEM Website and the “live” lube sheet. This living document is updated as the equip drives the job and allows the lube technician to plan his route based on what needs fuel and what doesn’t.
- This tablet has a software that can be used to pull the exact location of the piece of equipment needing fuel, oils or other lube services in the field.
 - o Allows access to the Job Equipment Management system for any information that may be needed while in the field.

o Fuel Management – OEM Solutions (part of the Smart Lube truck)

- Displays list of equipment within 200 ft
- The meter hours are displayed for the selected unit
- Tracks time and amount fuel and oil dispensed to equipment
- Uploads quantities into the customer database
- Eliminates man hours of data entry and possible error in records



The OEM Computer Display Screen in the back of the Lube Truck

o Fuel Master System

- This system is utilized at the Fuel Farm to dispense fuel into the Lube Trucks
- Eliminates man hours of data entry and record keeping
- Utilizes RFID technology to control fuel being dispensed, fuel will not be dispensed without the RFID tag
- Records fuel dispensed, tied into the ECM which checks for error codes and allows for the reduction of stolen fuel
- Enhances fuel control and accurate fuel records

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5.0 Resources Required

To implement these strategies there needs to be a dedicated electronics data 'champion' within the fleet department that is capable of evaluating the equipment fleet and deciding what pieces and/or systems fit best with the operation. A thorough vetting process should take place when looking at all of the products available on the market. This individual would also be responsible for creating and delivering the training to the entire equipment staff to ensure a clear consistent message is delivered. A dedicated Information Technology person would also be very beneficial to ensuring that systems will communicate effectively with each other before choosing vendor(s).

Once the strategy is deployed and systems are in place, each of the disciplines within the equipment department should have a dedicated person to monitor the data, run reports, and compile action items based on the incoming data. If the systems are left as part someone else's already long list of daily activities it may quickly get left behind and the true value will not be realized.

6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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